SECTION 1 — MODEL DESCRIPTION

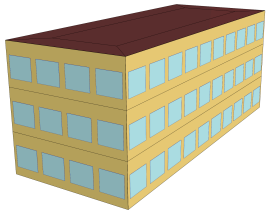
BUILDING DESCRIPTION

Sector	Commercial-Institutional
Building type	Office
Building subtype	Low-rise (1-3 storeys)
Vintage	Before 1945

HVAC SYSTEM

Heating	Electric boiler (hot water) + Hydronic baseboards / radiators
Cooling	Water-cooled chiller
Air distribution	Constant air volume (CAV) (1 air loop)
Heat recovery	None

3D MODEL



Three-dimensional geometric representation of the building model used for energy simulation.

BUILDING CHARACTERISTICS

Total floor area [m²]	1,040
Conditioned floor area [m²]	1,040
Above-ground storeys	3
Climate zone	6A
Weather file	Montreal-Trudeau Intl AP QC CAN CWEC2016 WMO#=716270

WINDOW-TO-WALL RATIO [%]

Average	40.0
North	40.0
East	40.0
South	40.0
West	40.0

ENVELOPE & INTERNAL LOADS

INSULATION & AIR TIGHTNESS

Exterior walls [W/m²K]	0.38
Roof [W/m²K]	0.41
Exposed floors / slabs [W/m²K]	2.94
Infiltration [m³/h/m² ext. surface]	0.82

GLAZING

Glazing U-factor [W/m²K]	2.96
SHGC	0.40

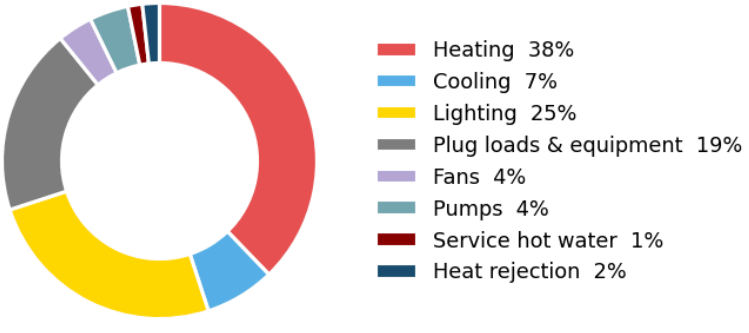
INTERNAL LOADS (MODEL INPUTS)

Occupant density [m²/person]	14.0
Lighting power density [W/m²]	6.7
Equipment power density [W/m²]	7.5

ANNUAL END USES [kWh/year]

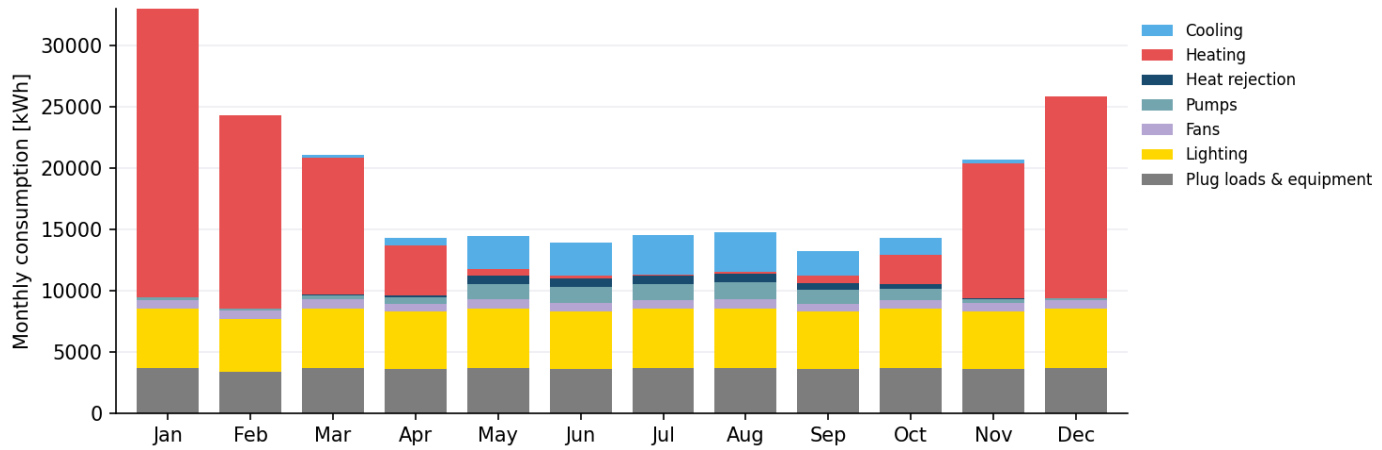
END USE	ELECTRICITY	DISTRICT HEATING	TOTAL	kWh/m²
Heating	86,136	0	86,136	82.8
Cooling	16,411	0	16,411	15.8
Interior lighting	57,050	0	57,050	54.9
Plug loads & equipment	43,686	0	43,686	42.0
Fans	8,247	0	8,247	7.9
Pumps	9,122	0	9,122	8.8
Heat rejection	3,981	0	3,981	3.8
Service hot water	0	3,386	3,386	3.3
Total			**228,019**	**219.3**

SHARE OF ANNUAL CONSUMPTION



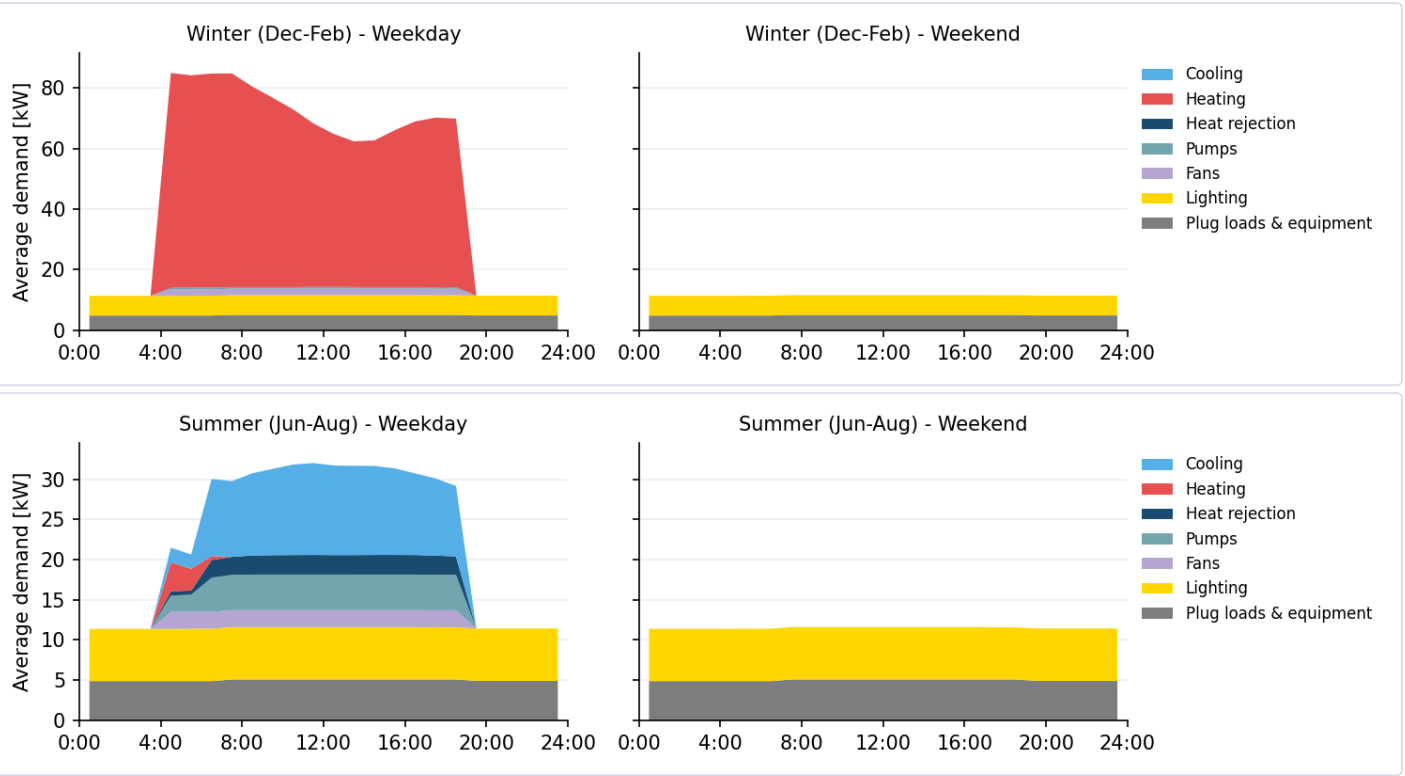
ENERGY USE INTENSITY	TOTAL CONSUMPTION	ELECTRICITY	NATURAL GAS	OTHER FUELS	PEAK ELECTRIC DEMAND
219 kWh/m²/yr (total area)	228,019 kWh/yr	224,633 kWh/yr	0 kWh/yr	3,386 kWh/yr	114 kW · 109.2 W/m²

MONTHLY CONSUMPTION BY END USE



Monthly energy consumption by end use, all fuels combined.

TYPICAL DAY PROFILES — WEEKDAY VS. WEEKEND



Average hourly demand by end use for winter and summer. Weekdays (Monday to Friday) are shown on the left and weekends and holidays on the right, showing the office occupancy schedule and HVAC set-back.

NOTES & ASSUMPTIONS

- Archetype energy model simulated with OpenStudio/EnergyPlus. Values are model outputs for the reference weather year and are not measured data.
- Energy use intensity is normalized by the total floor area (1,040 m²). Offices are not normalized per unit.
- Archetype: Low-rise office (1-3 storeys), built before 1945.
- HVAC entries marked "(typical)" are not modelled explicitly and come from the archetype assumption table.
- Peak electric demand occurs on 16-JAN-05:30.